

Causality Analysis Between Collaboration Profiles and Musical Success

Mariana O. Silva
mariana.santos@dcc.ufmg.br
Department of Computer Science, UFMG
Belo Horizonte, Brazil

Mirella M. Moro
mirella@dcc.ufmg.br
Department of Computer Science, UFMG
Belo Horizonte, Brazil

ABSTRACT

Current research in music collaboration networks has established that social interaction can impact the music industry's trajectory. In a previous study, we found that the way artists connect professionally may affect their musical success. Subsequently, our goal is to deepen such analyses by using time series and Granger causality test for assessing whether there is a causal relationship between collaboration profiles and artist's popularity. Our experimental evaluation reveals varied relationships linking collaboration profiles and musical success, indicating a direct connection between these two factors analyzed.

CCS CONCEPTS

- **Human-centered computing** → Collaborative and social computing;
- **Computing methodologies** → Machine learning;
- **Mathematics of computing** → Probability and statistics.

KEYWORDS

collaboration profiles, Granger causality, musical success, creative collaboration, ego networks

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1 INTRODUCTION

Artist collaboration is frequently thought as one of the vastest creative forces driving music today. Thanks to digital media, a broader variety of musical collaborations are emerging among artists who can often live on different continents. In fact, analyzing data from the Billboard Hot 100 (Figure 1), in the 1960s, 70s, and 80s, collaborations were relatively rare (about 5% of charted songs) and generally took the form of duets. The boom in collaborations started in the mid-1990s, when the number of collaborations increased significantly, with the duets dying and the artists on display taking over. Although such associations help artists bridge the gap between

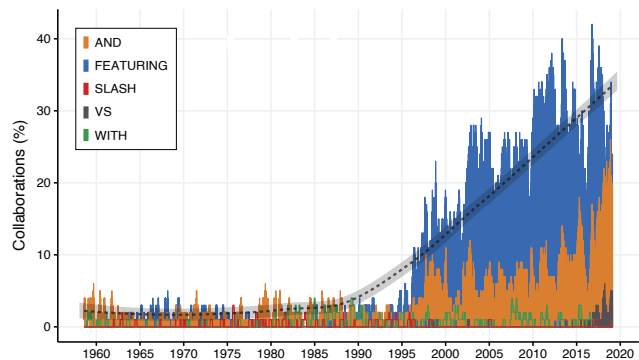


Figure 1: Billboard Hot 100 songs (1958 - 2019). Solid lines represent different types of musical collaborations. The dashed line indicates the trend of all collaborations.

styles and genres then crossing over to new fan bases, they often run the risk of sounding artificial and non authentic as defined by professional critics. Nevertheless, there is no denying that collaborations are on their way to conquering the popular music domain.

Despite the propagation of musical collaborations, there is limited knowledge about how these associations form and what factors lead to their success [13]. Previous research in music collaboration networks has established that social interaction can influence the trajectory of the music industry [2, 5, 10, 16, 21, 24]. In particular, Budner and Grahl [4] found that different collaborative roles hold distinct impacts on network connectivity. Also, the influence of a specific function depends on both the number of relationships and the importance of these connections. Overall, there is a need for a sounder understanding on how artists connect professionally and the impact of such connections on their music success.

Indeed, the ability to predict musical success offers tremendous benefits for many domains and audiences. For music industry CEOs, it can help maximize expected success by helping to decide whom to invest in to produce potential hits. Furthermore, by properly investing in potential artist/music and distribution, the studio could increase sales of both physical and digital albums, improve revenue from on-demand audio streaming services or even launch the next *it* artist or *summer hit*. Artists may also profit by identifying the most suitable songs to lead the album to early stardom. Even we, as researchers, may develop new tools and practices that help the music industry and its members flourish. Finally, for music consumers, it may help to decide if an album is worth buying because it may potentially contain 3-5 hits, instead of being a *one-hit-wonders* album. This ability is the intrinsic motivation for a new field of research

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known as Hit song science (HSS), which Pachet and Sony [17] define as “an emerging field of investigation that aims at predicting the success of songs before they are released on the market”.

The broad range of characteristics that lead to the success of a song exceeds its intrinsic content, namely the audio features and the lyrics. Factors such as the artist’s preferred attachment [3], society and culture [7] or psychological parameters on the reasons for preferring a track and listening exposure to tracks [17], can also fulfill a key role. Following an alternate direction, in a previous study [20], we argued that the way artists connect professionally can significantly affect musical success. Using data from Billboard and Spotify, we have identified three collaboration profiles in a musical network composed of 354 successful artists: (i) *Diverse Collaboration*, (ii) *Regular Collaboration* and (iii) *Absent*. Through a statistical analysis, we observed that the most successful artists are more likely to have highly collaborative profiles among influential and diverse artists (belonging to the *Diverse* profile). Contrariwise, those who prefer to pursue a non-collaborative music career (belonging to the *Absent* profile) may be missing an opportunity to improve and expand their potential.

Though the correlation between collaboration profiles and musical success is well established, there is no research into the causality in such relationship. Understanding both statistical terms (together) is very important, not only for drawing conclusions but more importantly, for a correct conclusion at the end. However, determining causation is never perfect in the real world and there are different methods to find evidence on causal relationships, including the Granger causality¹ test. Overall, we explore the potential of Granger causality [11] to assess the existence of a causal relation between collaboration profiles and musical success. Our contributions are summarized as follows.

- We explore the causal relationship between collaborative profiles and artist’s success;
- We conduct two analyses in parallel based on daily time series of collaborative songs and musical success, and solo songs and musical success;
- We validate that artistic popularity strongly affects establishing musical collaborations in a diverse profile;
- We confirm the presence of a vicious circle between musical collaborations and musical success in the most collaborative profiles; and
- We also confirm the *Absent* profile as the one with less successful artists, precisely because it is a non-collaborative profile.

The remainder of this paper is organized as follows. In Section 2, we discuss related work. In Section 3, we describe the dataset and introduce the proposed methodology. We detail the results and experimental evaluation in Section 4. We conclude with future research directions in Section 5.

2 RELATED WORK

Investigating and predicting creative collaborations’ success has been the subject of active research in the last decade [1, 2, 5, 6, 16]. In the context of musical success, the main starting point to

explain it from social network analysis was the study by Gleiser and Danon [10]. Using a dataset of jazz recordings, the authors investigated the impact of the collaboration network topology of jazz musicians and their social interactions. Following a similar line of thought, Uzzi and Spiro [24] conducted a social network analysis on a creative collaboration network composed by Broadway musical artists. Applying statistical methods, they observed that network measures significantly affected creativity in terms of financial and creative success.

In another setting, Budner and Grahl [4] analyzed the large-scale structure of musical collaborations by applying tools of network science. The authors found that different collaborative roles hold a distinct impact on network connectivity. Moreover, the influence of a specific collaborator role is not only dependent on the number of connections of the role but also on the importance of such connections. More recently, Calefato et al. [5] proposed a study on creative collaboration in a music community where authors compose songs together by mixing a new track with an existing audio recording (*overdubbing*). They analyzed the relationship between metrics related to the song and to the authors and the likelihood of a song being overdubbed.

Nevertheless, such papers ignore the hypothesis that the popularity of artists may be directly related to their *collaborative patterns*. In fact, most existing proposals are limited to analyzing non-topological metrics referring to the music or the artist, as well as topological measures related only to the nodes’ connections. Therefore, to further our understanding of how music collaboration profiles affect artists’ popularity, in a previous study [20], we analyzed and identified such characterizations in a musical network composed only of successful artists. Using data from Billboard and Spotify, we applied a clustering algorithm to detect groups of artists with similar collaboration profiles. In this process, we have identified three clusters with predominant collaboration profiles:

- **Diverse** composed by artists with a high degree of interaction, distant collaborations, and an average level of connections between different genres and musical collaborations between important artists.
- **Regular** composed by not very collaborative artists, with a high degree of distant collaboration and between the same musical genres, and rare musical collaborations among important artists.
- **Absent** composed of non-collaborative artists.

By applying a statistical analysis, we identified moderate and statistically significant correlations between popularity and measures of degree and centrality. Nonetheless, such previous study established a correlation between collaboration profiles and musical success. However, correlation is a statistical technique that evaluates only the intensity at which a pair of variables is linearly related. That is, having a direct mathematical correlation between variables does not necessarily mean that one causes the other. Therefore, the present study differs from all others, to the best of our knowledge, by being the first to model this correlation as a form of causal dependence between musical success and collaborations.

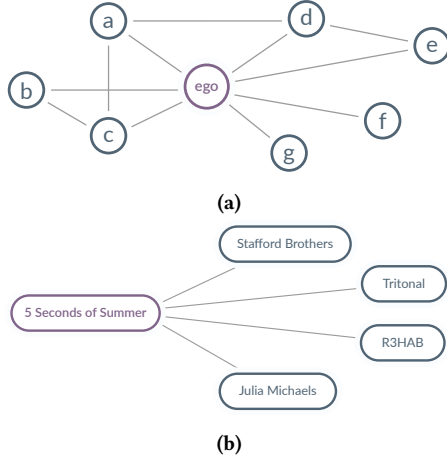
3 METHODOLOGY

Our methodology proposes investigating if there is a causal relationship between collaboration profiles and musical success. First, in

¹ *Granger’s causality* is a statistical concept of cause-and-effect relationship that is based on prediction. Note: it is different than the term *causality* in other contexts such as Pearl Causal Model (PCM) [18]

Table 1: Statistics of the top 30 artists selected

	Artist	# Solo	# Collabs		Artist	# Solo	# Collabs		Artist	# Solo	# Collabs
Diverse	Carlos Vives	188	71	Regular	XXXTENTACION	61	32	Absent	5 Seconds of Summer	186	4
	Drake	142	97		Ariana Grande	88	77		Adele	52	2
	Future	202	134		Bad Bunny	22	57		Arctic Monkeys	131	0
	J Balvin	91	141		Ed Sheeran	170	57		One Direction	125	17
	Kanye West	124	89		Eminem	333	150		Panic! At The Disco	172	8
	Kendrick Lamar	111	70		Khalid	42	38		Queen	893	72
	Nicki Minaj	188	170		Marshmello	29	75		Red Hot Chili Peppers	438	0
	Travis Scott	45	32		Nicky Jam	102	112		Sebastian Yatra	38	69
	Ty Dolla \$ign	44	167		Ozuna	33	71		The Beatles	825	14
	Wiz Khalifa	204	130		Post Malone	47	31		Twenty One Pilots	67	5

**Figure 2: (a) The ego network model: a top 30 artist, ego, and his/her collaborating artists, alter_{a-g}. (b) A real ego network example for the band 5 Seconds of Summer.**

Section 3.1, we describe the process of building a dataset containing 30 popular artists from three different collaboration profiles. From this data, we use ego networks to model such artists' collaborations in Section 3.2, and define a temporal success measure in Section 3.3. Next, we create three time series of musical success, musical collaborations and solo songs of the selected artists in Section 3.4. Finally, in Section 3.5, we apply the Granger causality test to check if patterns of collaboration can influence the musical success or vice versa.

3.1 Data Collection

In a previous study [20], we detected three collaboration profiles: Diverse, Regular and Absent. From these three clusters, we now select the 10 most popular artists in each profile. That is, in total, 30 successful artists were chosen. Using the Spotify library², we collected the entire discography of such artists. The *Spotipy* provides the Spotify catalog information about a certain artist's work, where four types of albums are available: *album*, *single*, *appears_on* and *compilation*. Here, we did not filter for a specific type, then considering all four types. Table 1 lists the names of all the artists present in our dataset³, as well as the total number of solo and collaborative songs of each one.

²Spotipy: <http://spotipy.readthedocs.io/en/latest/>

³Full dataset openly available at <https://homepages.dcc.ufmg.br/~mirella/projs/bade>

3.2 Ego Network Modeling

From the dataset, we created ego networks for each of the 30 selected artists. Ego networks have proved to be a valuable tool for understanding the relationships that individuals establish with their peers. In an ego network, the individual (termed *ego*) is at the center of the graph, and the edges connect her/his to the peers (termed *alters*) with whom she/he interacts [22]. As depicted in Figure 2, we model 30 ego networks, where the top 30 artists represent the ego nodes and their musical collaborations, the alter nodes.

3.3 Temporal Success Measure

Musical success can be defined through popularity on social media and music platforms, sales profit, awards, etc. Another common approach is to rely on pop charts, such as the Billboard charts⁴. In this case, the ranking of a song/album/artist in a chart is a time domain signal, describing its popularity over time. To define the temporal success measure, we first collected and grouped rankings of 18 top Billboard charts by the name of the top 30 artists. Next, we calculate the popularity over time of these artists from the *rank_score* [15] of the artists/albums/songs top charts. The *rank_score* is the inverted rating on a chart. That is, the *rank_score_{jt}(i)* of a chart *j* at time *t* of a song/album/artist *i* is obtained by Equation 1.

$$rank_score_{jt}(i) = max_rank_{jt} - rank_{jt}(i) + 1, \quad (1)$$

where *max_rank_{jt}* is the lowest rank of the *j* chart at time *t*, and *rank_{jt}(i)* is the rank of the song/album/artist *i*. For instance, in a weekly Top Album Sales chart, the album ranked highest has a *rank_score* of 100 and the album ranked lowest has a *rank_score* of 1. After calculating the *rank_scores* for each of the 18 charts for each of the artists over time, we aggregate these scores to obtain our temporal success measure, *S_t*. Then, we define the temporal success measure of an artist *i* with Equation 2.

$$S_t(i) = \sum_{j=1}^{18} rank_score_{jt}(i). \quad (2)$$

3.4 Time Series

Time series analysis is a powerful technique that can be used to understand the distinct temporal data patterns and to predict how levels of a variable will change in the future, considering what has happened in the past. To study the dynamic properties of the ego networks modeled in Section 3.2, we conducted two parallel

⁴Billboard charts: <https://www.billboard.com/charts>

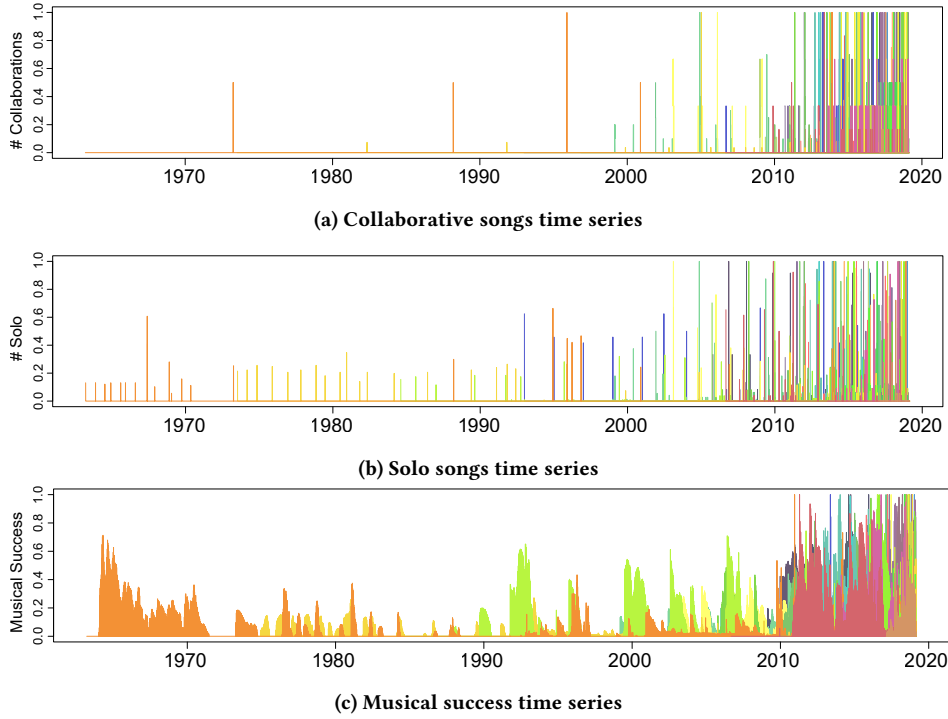


Figure 3: The three time series created for each of the 30 ego networks. Each color accordingly represents a top 30 artists time series. The time series vary from 1963 to 2019.

analyses based on daily time series of (i) collaborative songs and musical success and (ii) solo songs and musical success. Hence, in the end, each of the 30 ego networks produces three time series defined as follows.

- **Collaborative songs.** Daily time series containing the total number of musical collaborations normalized over time. The elapsed time of each of these series was collected from the release dates of all the artist's songs. See Figure 3a.
- **Solo songs.** Daily time series containing the total number of solo songs normalized over time. The elapsed time of each of these series was collected from the release dates of all the artist's songs. See Figure 3b.
- **Musical success.** Daily time series with the artist's success measure defined in Section 3.3 normalized over time. See Figure 3c.

3.5 Granger Causality Analysis

Finally, we analyze the effect of musical collaborations on musical success over time, applying the Granger causality test. Technically, the Granger causality test is a method to determine if one time series is useful in forecasting another. Since predictability remains a fundamental feature of causal attribution, Granger causality (GC) is also frequently interpreted in cause-effect context. However, GC should not be confused with the deep meaning of the word. GC is limited exclusively to identifying a statistical cause-and-effect relationship between two variables when there is a temporal precedence relationship between them.

The basic principle of the Granger causality test is to check if there is a lagged cause relationship between two or more variables. For example, we can investigate the causality between two time series of world prices of Arabica and Robusta coffees [23]. Let X_t and Y_t be the Arabica and Robusta coffees variables, respectively. X_t Granger-causes Y_t , if the forecast obtained in the current price of Robusta coffee can be improved by considering the lagged price information for Arabica coffee. In order to test the Granger causality, the series Y_t and X_t must be stationary, i.e., the series must be integrated with order zero $I(0)$. According to Enders [8], the following VAR (Vector Autoregression) model can be estimated to test the causality between two variables:

$$Y_t = \beta_1 + \beta_2 Y_{t-1} + \beta_3 X_{t-1} + \varepsilon_{1t} \quad (3)$$

$$X_t = \beta_4 + \beta_5 X_{t-1} + \beta_6 Y_{t-1} + \varepsilon_{2t}, \quad (4)$$

where ε_{1t} and ε_{2t} are white noises, (β_2, β_5) are the current and lagged coefficients of endogenous variables (determined within the model) and (β_3, β_6) are the current and lagged coefficients of exogenous variables (determined outside the model). These coefficients measure the influence of past values of each variable. In this system of equations, each variable is a function of its value with a lag and the value of the other variable with a lag. After verifying if all the variables are stationary, we can use the VAR model with n lags in Equations 5 and 6.

Table 2: Granger causality test for the first analysis. Column 1 groups the results by profile. Column 2 shows the selected top 30 artists. Columns 3 and 4 show the data characteristics of the series *COLLAB* (number of musical collaborations) and *POP* (musical success) respectively. If the series itself is stationary, we represent it by $I(0)$. If the series is integrated with order n , we represent it by $I(n)$. Column 5 shows the lag orders, where the lag order chosen is underlined. Column 6 uses $\checkmark$ to indicate that the two series are co-integrated. Columns 7 and 8 represent the P-value of the Granger causality test: *, **, * and . statistically significant at 0%, 1%, 5% and 10%, respectively**

	Artist	<i>COLLAB</i>	<i>POP</i>	Lag	Co-integration	<i>COLLAB</i> $\rightarrow$ <i>POP</i>	<i>POP</i> $\rightarrow$ <i>COLLAB</i>
Diverse	Drake	$I(0)$	$I(1)$	15 7		5.756e-07 ***	0.9831
	J Balvin	$I(1)$	$I(1)$	20 14	$\checkmark$	4.988e-06 ***	4.988e-06 ***
	Carlos Vives	$I(1)$	$I(1)$	17 15	$\checkmark$	0.2477	< 2.2e-16 ***
	Nicki Minaj	$I(0)$	$I(1)$	14		0.032 *	0.9881
	Kanye West	$I(0)$	$I(1)$	56 14 7		< 2.2e-16 ***	1
	Kendrick Lamar	$I(0)$	$I(1)$	14 10 9		5.827e-1 ***	2.571e-06 ***
	Future	$I(0)$	$I(1)$	61 7		< 2.2e-16 ***	< 2.2e-16 ***
	Ty Dolla \$ign	$I(0)$	$I(1)$	14 7		0.491 *	0.5503 **
	Travis Scott	$I(0)$	$I(1)$	21 14 7		0.3772	0.7933
	Wiz Khalifa	$I(0)$	$I(1)$	20 14		6.872e-10 ***	6.872e-10 ***
Regular	XXXTENTACION	$I(0)$	$I(1)$	7		0.6671	0.7559
	Post Malone	$I(0)$	$I(1)$	14 7		0.2291	5.379e-11 ***
	Ozuna	$I(1)$	$I(1)$	20 14	$\checkmark$	0.000236 ***	0.003717 **
	Bad Bunny	$I(1)$	$I(1)$	7	$\checkmark$	0.2568	0.994
	Khalid	$I(1)$	$I(1)$	17 14	$\checkmark$	4.359e-07 ***	3.546e-14 ***
	Ed Sheeran	$I(1)$	$I(1)$	19 17 14	$\checkmark$	1.371e-11 ***	< 2.2e-16 ***
	Ariana Grande	$I(0)$	$I(1)$	7		0.6814	0.0009666 ***
	Nicky Jam	$I(1)$	$I(1)$	99 59 15 14	$\checkmark$	< 2.2e-16 ***	< 2.2e-16 ***
	Eminem	$I(0)$	$I(1)$	54 14 7		3.344e-15 ***	1
	Marshmello	$I(1)$	$I(1)$	97 21 14 7	$\checkmark$	0.112	8.122e-07 ***
Absent	Panic! At The Disco	$I(1)$	$I(1)$	14 7	$\checkmark$	< 2.2e-16 ***	0.1697
	5 Seconds of Summer	$I(1)$	$I(1)$	7	$\checkmark$	0.3544	5.002e-15 ***
	Twenty One Pilots	$I(0)$	$I(1)$	7		0.0001239 ***	0.9952
	The Beatles	$I(0)$	$I(1)$	20 17 14		1	1.711e-14 ***
	Queen	$I(1)$	$I(1)$	98 21 14 7	$\checkmark$	0.9869	0.9922
	Red Hot Chili Peppers	$I(0)$	$I(1)$	1		-	-
	Arctic Monkeys	$I(0)$	$I(1)$	20 15 14		0.7243	2.728e-12 ***
	Sebastian Yatra	$I(1)$	$I(1)$	16 8 7	$\checkmark$	6.852e-11 ***	0.722
	Adele	$I(0)$	$I(1)$	20 15 14		0.7243	2.728e-12 ***
	One Direction	$I(0)$	$I(1)$	36 7		0.001496 **	2.043e-05 ***

$$Y_t = \sum_{i=1}^n \alpha_i Y_{t-i} + \sum_{j=1}^n \beta_j X_{t-j} + \varepsilon_{1t} \quad (5)$$

$$X_t = \sum_{k=1}^n \gamma_k Y_{t-k} + \sum_{l=1}^n \delta_l X_{t-l} + \varepsilon_{2t} \quad (6)$$

Equation 5 postulates that current values of Y_t are related to past values of Y_t as well as to lagged values of X_t . On the other hand, Equation 6 postulates a similar behavior for the variable X_t . In general terms, since the future can not predict the past, if the variable X_t *Granger-causes* the variable Y_t , then changes in X_t must temporarily precede changes in Y_t . In order to test whether X_t *Granger-causes* Y_t , we attempt to reject or accept the following hypotheses by means of F-tests as follows:

$$H_0 : \sum_{j=1}^n \beta_j = 0$$

$$H_1 : \beta_j \neq 0, \text{ for at least one } j,$$

where j is the number of lags. Hence, the null hypothesis (H_0) in which X_t does not *Granger-cause* Y_t is tested against the alternative hypothesis (H_1) that at least one lag of the variable X_t *Granger-causes* Y_t . Based on the estimated coefficients for Equations 5 and 6, four different hypotheses about the relationship between Y_t and X_t can be formulated, as follows.

Unidirectional Granger-causality $X_t \rightarrow Y_t$: when the estimated coefficients of the lags of X_t in Equation 5 are jointly different from zero ($\sum_{j=1}^n \beta_j \neq 0$), and the estimated coefficients of the lags of Y_t in Equation 6 are jointly zero ($\sum_{k=1}^n \gamma_k = 0$).

Unidirectional Granger-causality $Y_t \rightarrow X_t$: when the estimated coefficients of the lags of X_t in Equation 5 are jointly zero ($\sum_{j=1}^n \beta_j = 0$), and the estimated coefficients of the lags of Y_t in Equation 6 are jointly different from zero ($\sum_{k=1}^n \gamma_k \neq 0$).

Bidirectional (or feedback) Granger-causality: when the sets of lagged coefficients of X_t and Y_t are statistically different from zero in both regressions ($\sum_{j=1}^n \beta_j \neq 0$ and $\sum_{k=1}^n \gamma_k \neq 0$).

Independence between X_t and Y_t : when, in both regressions, the sets of lagged coefficients of X_t and Y_t are not statistically different from zero ($\sum_{j=1}^n \beta_j = 0$ and $\sum_{k=1}^n \gamma_k = 0$).

4 EVALUATIONS AND DISCUSSION

This section goes over our analyses through the times series evaluation and a discussion over the results.

4.1 Times Series Evaluation

Following Toda and Yamamoto [23], the first step in the Granger causality test is to verify if the time series are stationary. Hence, we applied the Augmented Dickey-Fuller (ADF) [9] and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) [14] tests. For both time series, the

Table 3: Granger causality test for the second analysis. Column 1 groups the results by profile. Column 2 shows the selected top 30 artists. Columns 3 and 4 show the data characteristics of the series *SOLO* and *POP* series respectively. If the series itself is stationary, we represent it by $I(0)$. If the series is integrated with order n , we represent it by $I(n)$. Column 5 shows the lag orders, where the lag order chosen is underlined. Column 6 uses $\checkmark$ to indicate that the two series are co-integrated. Columns 7 and 7 represent the P-value of the Granger causality test: *, **, * and . statistically significant at 0%, 1%, 5% and 10%, respectively**

	Artist	<i>SOLO</i>	<i>POP</i>	Lag	Co-integration	<i>SOLO</i> $\rightarrow$ <i>POP</i>	<i>POP</i> $\rightarrow$ <i>SOLO</i>
Diverse	Drake	$I(0)$	$I(1)$	<u>15</u>		0.121	< 2.2e-16 ***
	J Balvin	$I(0)$	$I(1)$	<u>15</u>		0.3623	< 2.2e-16 ***
	Carlos Vives	$I(0)$	$I(1)$	<u>21</u>		0.9976	< 2.2e-16 ***
	Nicki Minaj	$I(0)$	$I(1)$	<u>19</u> 7		0.4811	< 2.2e-16 ***
	Kanye West	$I(0)$	$I(1)$	<u>15</u> 7		0.1103	1.671e-1 ***
	Kendrick Lamar	$I(0)$	$I(1)$	<u>22</u> 7		0.477	< 2.2e-16 ***
	Future	$I(0)$	$I(1)$	<u>7</u>		0.4268	0.9981
	Ty Dolla \$ign	$I(0)$	$I(1)$	<u>22</u> 15 7		2.403e-09 ***	< 2.2e-16 ***
	Travis Scott	$I(0)$	$I(1)$	<u>15</u>		1	< 2.2e-16 ***
	Wiz Khalifa	$I(0)$	$I(1)$	<u>18</u> 7		0.8481	< 2.2e-16 ***
Regular	XXXTENTACION	$I(0)$	$I(1)$	<u>15</u> 7		0.766	< 2.2e-16 ***
	Post Malone	$I(0)$	$I(1)$	<u>15</u>		0.9966	< 2.2e-16 ***
	Ozuna	$I(0)$	$I(1)$	<u>15</u>		0.7705	< 2.2e-16 ***
	Bad Bunny	$I(0)$	$I(1)$	98 71 <u>65</u> 20 13 7		< 2.2e-16 ***	< 2.2e-16 ***
	Khalid	$I(1)$	$I(1)$	<u>17</u> <u>16</u>	$\checkmark$	2.735e-05 ***	< 2.2e-16 ***
	Ed Sheeran	$I(0)$	$I(1)$	<u>22</u> 14		0.872	2.147e-14 ***
	Ariana Grande	$I(0)$	$I(1)$	<u>22</u> <u>15</u> 7		0.2358	< 2.2e-16 ***
	Nicky Jam	$I(1)$	$I(1)$	<u>29</u> <u>22</u>	$\checkmark$	1	< 2.2e-16 ***
	Eminem	$I(0)$	$I(1)$	<u>20</u> <u>19</u> 7		0.9331	< 2.2e-16 ***
	Marshmello	$I(0)$	$I(1)$	<u>14</u> <u>7</u>		0.05908 .	0.2674
Absent	Panic! At The Disco	$I(0)$	$I(1)$	<u>22</u> 7		0.9999	< 2.2e-16 ***
	5 Seconds of Summer	$I(0)$	$I(1)$	<u>7</u>		0.9876	1
	Twenty One Pilots	$I(0)$	$I(1)$	<u>15</u>		0.9999	< 2.2e-16 ***
	The Beatles	$I(0)$	$I(1)$	38 <u>22</u> 8 7		0.6306	< 2.2e-16 ***
	Queen	$I(0)$	$I(1)$	98 <u>22</u> 7		0.999	1.697e-07 ***
	Red Hot Chili Peppers	$I(0)$	$I(1)$	22 <u>21</u> 14 7		1	< 2.2e-16 ***
	Arctic Monkeys	$I(0)$	$I(1)$	<u>52</u> 35 7		0.4089	4.898e-08 ***
	Sebastian Yatra	$I(0)$	$I(1)$	<u>15</u>		3.355e-05 ***	< 2.2e-16 ***
	Adele	$I(0)$	$I(1)$	<u>52</u> 35 7		0.4089	4.898e-08 ***
	One Direction	$I(0)$	$I(1)$	<u>19</u> 7		1.57e-05 ***	< 2.2e-16 ***

null hypothesis H_0 of non-stationary can be rejected at the 5% significance level. Tables 2 and 3 present, respectively, the results for ADF and KPSS to verify the order of integration of the level series for the two analyses. According to the results, all *POP* series are integrated of order one $I(1)$ at a significance level of 5%. However, the great majority of *COLLAB* and *SOLO* series are stationary with $I(0)$ at a significance level of 5%. For all the time series group, we consider the maximum integration order $m = 1$.

In cases where both series have the same integration order, the co-integration test must be applied between the variables. Therefore, we performed the Johansen co-integration test for the 14 pairs of time series that presented the same integration order (See Tables 2 and 3). As highlighted by Toda and Yamamoto, the lag selection procedure is a crucial step for the Granger causality test. In this methodology, in order to determine the number of lags, p , we first estimate an unrestricted VAR(p) model and select the model that minimized the information criteria of Akaike (AIC), Hannan-Quinn (HQ) [12] and Schwarz (SC) [19].

Based on Tables 2 and 3 and following Toda and Yamamoto, we added to the VAR(p) models, additional m lags in each of the variables of each of the equations – where p is the appropriate maximum lag for the variables, and m represents the maximum integration order for the time series. Next, we test the Granger causality as follows. For expository purposes, suppose VAR(p) has two equations, for X_t and for Y_t . We test the hypothesis that (only) the coefficients of the first lagged values of X_t are zero in equation Y_t , using the F-test. Subsequently, we perform the same analysis for the coefficients of the lagged values of Y_t in equation X_t .

Hypotheses. As mentioned earlier, we conducted two parallel analyses based on daily time series of collaborative songs and musical success, and solo songs and musical success, from each of the top 30 artists selected from the three collaboration profiles (*Diverse*, *Regular* and *Absent*). In both analyses, we test the null hypothesis H_0 that collaborative profiles do not *Granger-cause* artistic success. From the results of both analyses, we can distinguish four hypotheses as follows.

Hypothesis 1. The way in which artists professionally connect with each other is useful in forecasting their musical success. This hypothesis imagines a situation in which an artist's collaborative patterns have a direct impact on his/her success.

Hypothesis 2. The musical success of an artist is useful in forecasting his/her collaborative profile. This hypothesis imagines a situation in which the musical success of artists drives the way in which they collaborate.

Hypothesis 3. The musical success of an artist assists in forecasting their collaborative profile, as well as their profile helps in forecasting their success. This hypothesis imagines a situation in which artists, before collaborating with other people, they take into consideration the success level of these individuals. In the same way, the growth of artistic success is affected by the way artists collaborate professionally.

Hypothesis 4. There is no causality relationship (in the Granger sense) between collaboration profiles and musical success.

Table 4: Summary of main results (Tables 2 and 3) regarding our four hypotheses

COLLAB & POP				SOLO & POP			
Artists (%)		Causality Relation	Hypothesis	Artists (%)		Causality Result	Hypothesis
Diverse	40%	$COLLAB \leftrightarrow POP$	Hypothesis 3	Diverse	80%	$SOLO \rightarrow POP$	Hypothesis 1
	30%	–	Hypothesis 4		10%	$POP \leftrightarrow SOLO$	Hypothesis 3
	20%	$POP \rightarrow COLLAB$	Hypothesis 2		10%	–	Hypothesis 4
	10%	$COLLAB \rightarrow POP$	Hypothesis 1				
Regular	40%	$COLLAB \leftrightarrow POP$	Hypothesis 3	Regular	70%	$SOLO \rightarrow POP$	Hypothesis 1
	30%	$COLLAB \rightarrow POP$	Hypothesis 1		20%	$POP \leftrightarrow SOLO$	Hypothesis 3
	20%	–	Hypothesis 4		10%	–	Hypothesis 4
	10%	$POP \rightarrow COLLAB$	Hypothesis 2				
Absent	50%	$COLLAB \rightarrow POP$	Hypothesis 1	Absent	70%	$SOLO \rightarrow POP$	Hypothesis 1
	30%	$POP \rightarrow COLLAB$	Hypothesis 2		20%	$POP \leftrightarrow SOLO$	Hypothesis 3
	20%	–	Hypothesis 4		10%	–	Hypothesis 4

Hypotheses Summary

1. The way in which artists professionally connect with each other is useful in forecasting their musical success.
2. The musical success of an artist is useful in forecasting his/her collaborative profile.
3. The musical success of artists assists in forecasting their collaborative profile, as well as their profile helps in forecasting their success.
4. There is no causality relationship (in the Granger sense) between collaboration profiles and musical success.

4.2 Discussion

For clarification purposes, we start our discussion by summarizing the three base profiles [20]: *Diverse* has highly collaborative and important artists; *Regular* has collaborative, regular artists; and *Absent* has non-collaborative artists. Now, we further discuss the results from three perspectives: the collaborative songs, the solo ones, and some relevant general points crossing all information with the three profiles.

Collaborative Songs. Table 2 suggests that in 40% of *Diverse* profiles, musical collaboration helps predicting artistic success, just as success helps in predicting collaborations. In 20% of cases (*Drake* and *Kayne West*), there is unilateral causality where the success of the artist strongly influences the incidence of their musical collaborations. Only one artist (*Carlos Vives*) presented an opposite result, where his musical associations directly affect his success. The other 30% did not present statistically significant results.

Likewise, for 40% of the *Regular* profile, musical collaboration helps predicting artistic success, just as success helps predict collaborations. Now, in 30% of the cases (*Ariana Grande*, *Marshmello* and *Post Malone*), there is a unilateral causality, where, this time, the artist's musical associations directly affect its success. Only one artist (*Eminem*) has produced a divergent result, and his success affects the incidence of his musical collaborations. The other 20% did not present statistically significant results.

For the *Absent* profile, the tests indicate no bilateral causality. In fact, in 30% of cases (*Panic! At The Disco*, *Twenty One Pilots* and *Sebastian Yatra*), there is unilateral causality, where success helps predicting collaborations. In 50% of cases, there is a unilateral causality between the artist's musical associations and their success. The other 20% did not present statistically significant results.

Solo Songs. The results summarized in Table 3 suggest that, for *Diverse*, in 80% of the cases solo songs assists in predicting artistic success. This time, only one artist (*Ty Dolla \$ign*) presented bilateral causality, and the remaining 10% did not present statistically significant results. For the *Regular* profile, tests indicate that in 70% of the cases solo songs assists in predicting artistic success. Two artists (*Bad Bunny* and *Khalid*) presented a bilateral causality, and one artist reports no statistically significant results. In conclusion, for the *Absent* profile, the tests also indicate that in 70% of the cases solo songs assists in predicting artistic success. Two artists

(*Sebastian Yatra* and *One Direction*) presented a bilateral causality, and one artist indicates no statistically significant results.

General Points. Note that if two or more time series are cointegrated, then there must be *Granger causality* between them—either unidirectional or in both directions. Nevertheless, the opposite is false. According to Table 2, the time series of *Bad Bunny* and *Queen* are cointegrated, but no evidence of causality was found. In both cases, such a conflict may have occurred because the size of the sample is too small to satisfy the asymptotics on which the cointegration and causality tests depend. Similarly, the sample size of the *Red Hot Chili Peppers* series was also too small to perform the causality test.

Table 4 summarizes the final results of the Granger causality test for the two analyses considered: (i) percentage of artists, (ii) the test result and (iii) the confirmed hypothesis. Analyzing the two collaborative profiles (*Diverse* and *Regular*), when collaborating, most artists consider the level of success of their collaborators. Likewise, the growth of an artist's success is affected by the same musical collaborations. That is, the results indicate the presence of contemporary feedback between the two variables, forming a vicious circle between collaborations and musical success.

Regarding only the *Diverse* profile, there is a significant unilateral causal relationship between the solo songs and the artists' success, as well as a unilateral causal relationship between artistic success and musical collaborations. The first observation was predictable: as we selected only the most popular artists from each profile, it was expected that their solo songs would affect their musical success. However, the second observation shows that, contrary to expectations, the musical success of an artist affects the forecast of musical collaborations more than the opposite. Although it goes against our assumptions, such results still make sense. In our previous study outcomes [20], we found that *Diverse* was the most collaborative profile and composed of the most successful artists. Now, we can conclude that a large number of collaborations performed by the artists of this profile is because they are successful artists. That is, other musicians, seeking to increase their professional success, will always attempt to collaborate with the artists in such a group.

Regarding the *Regular* profile, the results also suggest the existence of a significant unilateral causal relationship between the solo songs and the artists' success. However, in this case, there is also a unilateral causal relationship between musical collaborations and artistic success. As in the *Diverse* profile, the first observation was

expected. Nevertheless, unlike the previous analysis, the second observation is consistent with our thesis. Being a profile composed of artists with success and an ordinary amount of collaborations, the eventual musical collaborations would cause (in Granger sense) the increasing of these artists' success.

Finally, regarding the *Absent* profile, the results suggest the existence of a significant unilateral causal relationship between the solo songs and the artists' success, once again. Moreover, for most artists, a unilateral causal relationship between musical collaborations and artistic success has been identified. As the *Absent* profile is composed of non-collaborative artists, this observation was less evident. However, through the current results, we can conclude that the few collaborations performed by these artists exposed a strong impact on their success. That is, as shown earlier [20], the reason these artists are the least successful (compared to the other profiles) may be related to this causal relationship. Yet in other words, because they collaborate very little, they may be missing the opportunity to improve and expand their musical success.

Limitations. Despite the relevant results, the dataset employed provides some constraints. By considering a limited number of successful artists, the data analysis risks becoming generic and biased. Therefore, including a more complete dataset, containing both popular and non-popular artists is certainly one of the following study steps. Another notable limitation is the questionable causal notion incorporated in Granger causality concept. According to Pearl [18], econometric concepts such as Granger causality are classified as statistical rather than causal. Hence, to further understanding the causal relationships between collaborative profiles and artist's success, we plan to deeply investigate techniques of the Pearl Causal Model (PCM).

5 CONCLUSION

In this paper, we selected and collected data from the 30 most successful artists from three different collaboration profiles. We then defined a temporal measure of musical success by grouping the *rank_score* of 18 Billboard charts. From our dataset, we created three time-series for each of the top 30 artists: collaborative songs, solo songs and musical success. With data from these time series and the success measure, we applied the Granger causality test to explore the causal relationship between collaborative profiles and artist's success. By conducting two analyses in parallel, the results for the causality test showed the causal relationship between the two variables is not obvious in general.

Combining the current results with observations from our previous study, we were able to understand more deeply the relationship between artists' collaborative patterns and their musical success. Specifically, we have evidence to support that artistic success strongly affects establishing musical collaborations in a *Diverse* profile. In addition, we confirmed the presence of a vicious circle between musical collaborations and musical success in the most collaborative profiles (*Diverse* and *Regular*). Finally, we verified *Absent* was the profile with less successful artists precisely because it is a non-collaborative profile.

As future work, we plan to tackle the two limiting factors of this study: a more complete dataset that includes "non-successful" artists

as well, and further analysis of causality using other approaches such as the Pearl Causal Model.

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REFERENCES

- [1] Carlos V.S. Araujo et al. 2017. Predicting Music Success Based on Users' Comments on Online Social Networks. In *Proceedings of the 23rd Brazilian Symposium on Multimedia and the Web*. Gramado, RS, Brazil, 149–156. <https://doi.org/10.1145/3126858.3126885>
- [2] Carlos V.S. Araujo et al. 2017. Using Complex Networks to Assess Collaboration in Rap Music: A Study Case of DJ Khaled. In *Proceedings of the 23rd Brazilian Symposium on Multimedia and the Web*. Gramado, RS, Brazil, 425–428. <https://doi.org/10.1145/3126858.3131605>
- [3] Kerstin Bischoff et al. 2009. Social Knowledge-Driven Music Hit Prediction. In *Advanced Data Mining and Applications*, Ronghuai Huang, Qiang Yang, Jian Pei, João Gama, Xiaofeng Meng, and Xue Li (Eds.). Berlin, Heidelberg, 43–54. https://doi.org/10.1007/978-3-642-03348-3_8
- [4] Pascal Budner and Jörn Grahl. 2016. Collaboration Networks in the Music Industry. *CoRR abs/1611.00377* (2016). <http://arxiv.org/abs/1611.00377>
- [5] Fabio Calefato et al. 2018. Collaboration Success Factors in an Online Music Community. In *Proceedings of the 2018 ACM Conference on Supporting Groupwork*. Sanibel Island, Florida, USA, 61–70. <https://doi.org/10.1145/3148330.3148346>
- [6] Tracy Cohen and Ben Clemens. 2005. Social networks for creative collaboration. In *Proceedings of the 5th conference on Creativity & cognition*. 252–255.
- [7] Ruth Dhanaraj and Beth Logan. 2005. Automatic Prediction of Hit Songs.. In *ISMIR*. 488–491.
- [8] Walter Enders. 2004. *Applied Econometric Times Series: And Rats Handbook for Econometric Time Series*, 2 Re. John Wiley & Sons.
- [9] Wayne A Fuller. 2009. *Introduction to statistical time series*. Vol. 428. John Wiley & Sons.
- [10] Pablo M Gleiser and Leon Danon. 2003. Community structure in jazz. *Advances in Complex Systems* 6, 04 (2003), 565–573. <https://doi.org/10.1142/S0219525903001067>
- [11] Clive WJ Granger. 1969. Investigating causal relations by econometric models and cross-spectral methods. *Econometrica: Journal of the Econometric Society* (1969), 424–438.
- [12] Edward J Hannan and Barry G Quinn. 1979. The determination of the order of an autoregression. *Journal of the Royal Statistical Society: Series B (Methodological)* 41, 2 (1979), 190–195.
- [13] Giuseppe Iaffaldano. 2018. Investigating Collaboration Within Online Communities: Software Development Vs. Artistic Creation. In *Proceedings of the 2018 ACM Conference on Supporting Groupwork*. 384–387.
- [14] Denis Kwiatkowski et al. 1992. Testing the null hypothesis of stationarity against the alternative of a unit root: How sure are we that economic time series have a unit root? *Journal of econometrics* 54, 1-3 (1992), 159–178.
- [15] Junghyuk Lee and Jong-Seok Lee. 2018. Music Popularity: Metrics, Characteristics, and Audio-Based Prediction. *IEEE Transactions on Multimedia* 20, 11 (2018), 3173–3182.
- [16] Siobhan McAndrew and Martin Everett. 2015. Music as Collective Invention: A Social Network Analysis of Composers. *Cultural Sociology* 9, 1 (2015), 56–80. <https://doi.org/10.1177/1749975514542486>
- [17] François Pachet and CSL Sony. 2012. Hit song science. *Music data mining* (2012), 305–326.
- [18] Judea Pearl. 2000. *Causality: models, reasoning and inference*. Vol. 29. Springer.
- [19] Gideon Schwarz et al. 1978. Estimating the dimension of a model. *The annals of statistics* 6, 2 (1978), 461–464.
- [20] Mariana O. Silva, Lais M. Rocha, and Mirella M. Moro. 2019. Collaboration Profiles and Their Impact on Musical Success. In *Proceedings of the 34th ACM/SIGAPP Symposium on Applied Computing*. Limassol, Cyprus, 2070–2077. <https://doi.org/10.1145/3297280.3297483>
- [21] Reginald D Smith. 2006. The network of collaboration among rappers and its community structure. *Journal of Statistical Mechanics: Theory and Experiment* 2006, 02 (2006), P02006.
- [22] Alistair Sutcliffe et al. 2012. Relationships and the social brain: integrating psychological and evolutionary perspectives. *British journal of psychology* 103, 2 (2012), 149–168.
- [23] Hiro Y Toda and Taku Yamamoto. 1995. Statistical inference in vector autoregressions with possibly integrated processes. *Journal of econometrics* 66, 1-2 (1995), 225–250.
- [24] Brian Uzzi and Jarrett Spiro. 2005. Collaboration and Creativity: The Small World Problem. *Amer. J. Sociology* 111, 2 (2005), 447–504. <https://doi.org/10.1086/432782>